LAB 7 – FUZZ TESTING USING HYPOTHESIS

Customs Duty Application

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Course: B.Tech – Computer Science and Engineering

1. Objective

To apply property-based fuzz testing with Hypothesis to the Customs Duty application, identify an input-handling bug, correct the parsing logic, and verify the application using automated tests.

2. Tools Used

Python 3.13, Hypothesis, Pytest, pytest-cov and Visual Studio Code.

3. Files Used

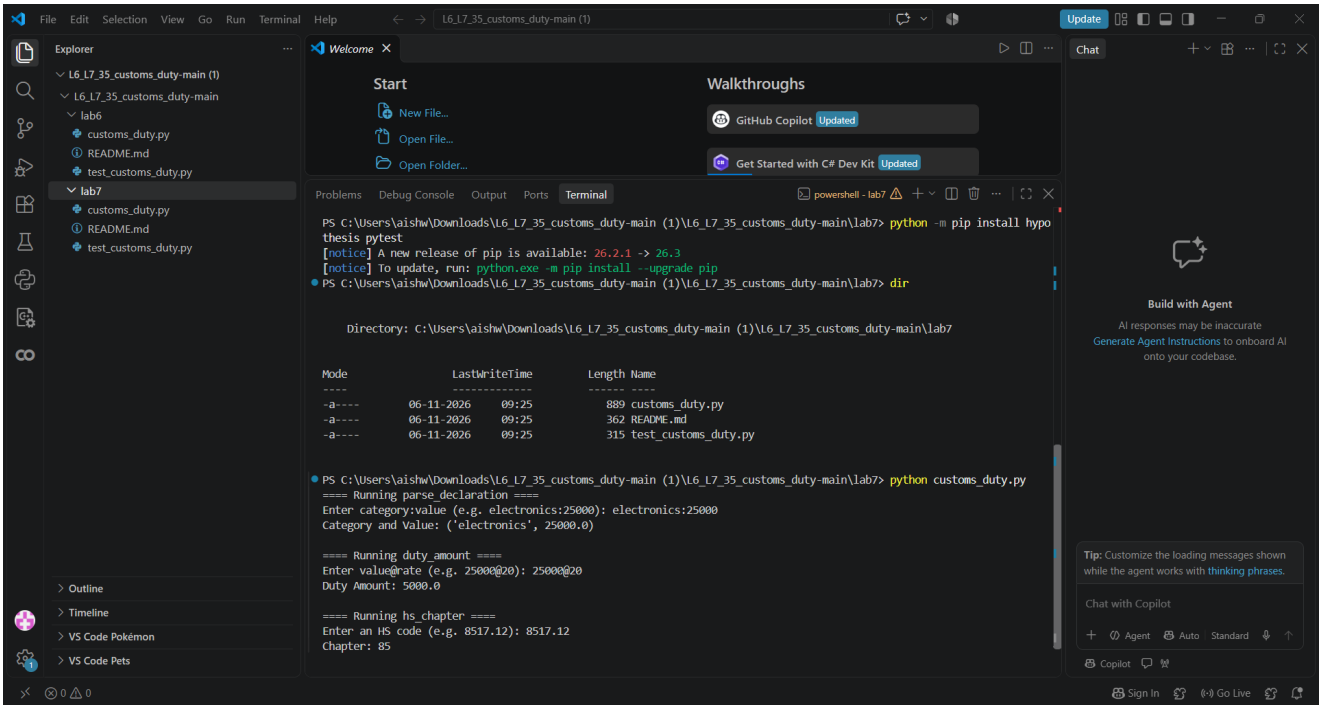
Application: customs_duty.py
Test file: test_customs_duty.py

4. Procedure

1. Installed Hypothesis and Pytest.
2. Ran the original customs_duty.py application with sample inputs.
3. Wrote Hypothesis tests for parse_declaration(), duty_amount() and hs_chapter().
4. Ran the tests and examined the failing input generated by Hypothesis.
5. Corrected parse_declaration() to split the input at the first colon only.
6. Re-ran the tests and checked the final test and coverage results.

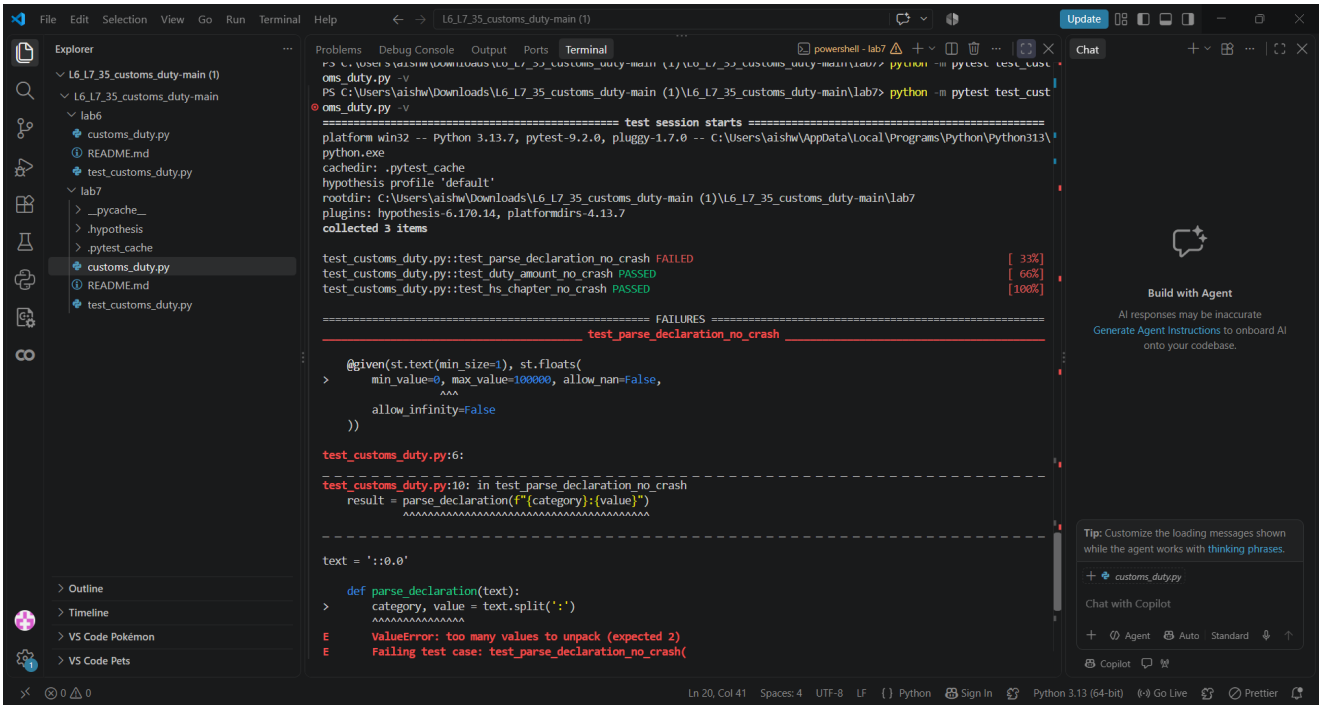
Screenshot 1 – Original Customs Duty application

The original application was executed with sample inputs for declaration parsing, duty calculation and HS chapter extraction.



Screenshot 2 – Initial Hypothesis failure

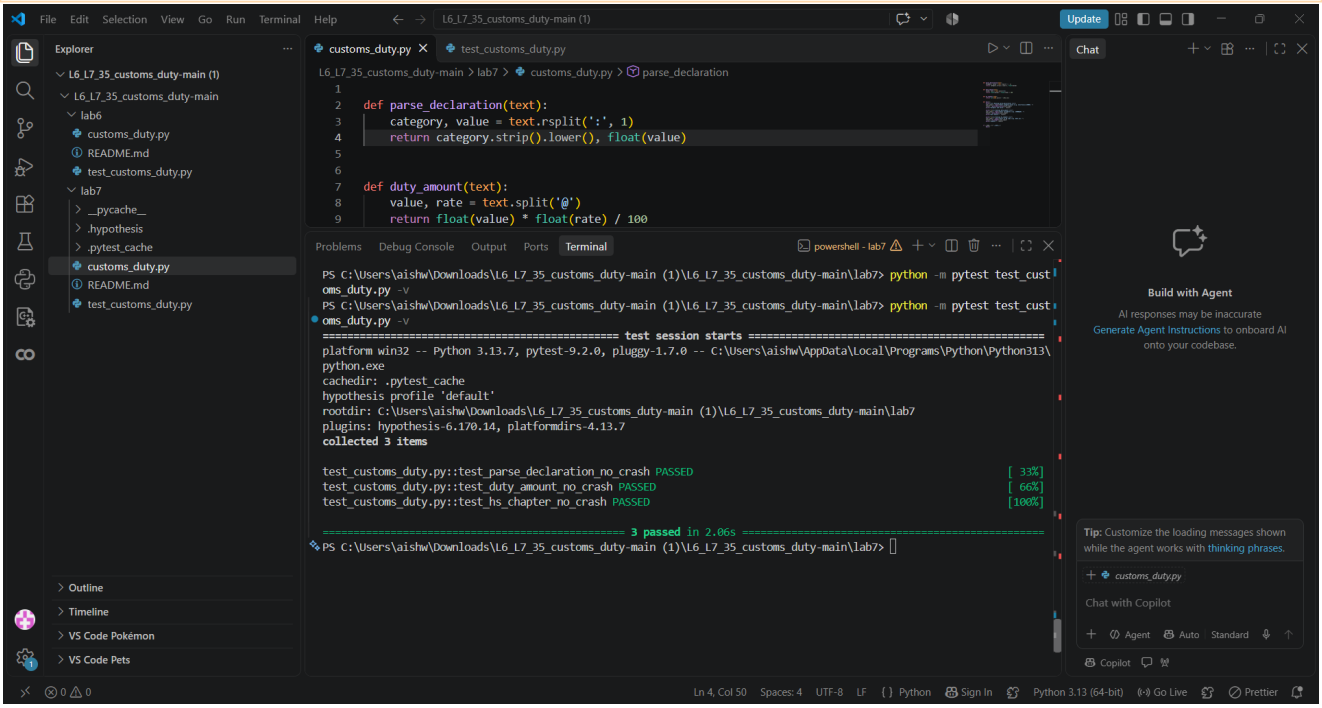
Hypothesis generated an input containing more than one colon. The original split operation attempted to unpack too many values and raised a ValueError.



Screenshot 3 – Corrected parsing logic and initial retest

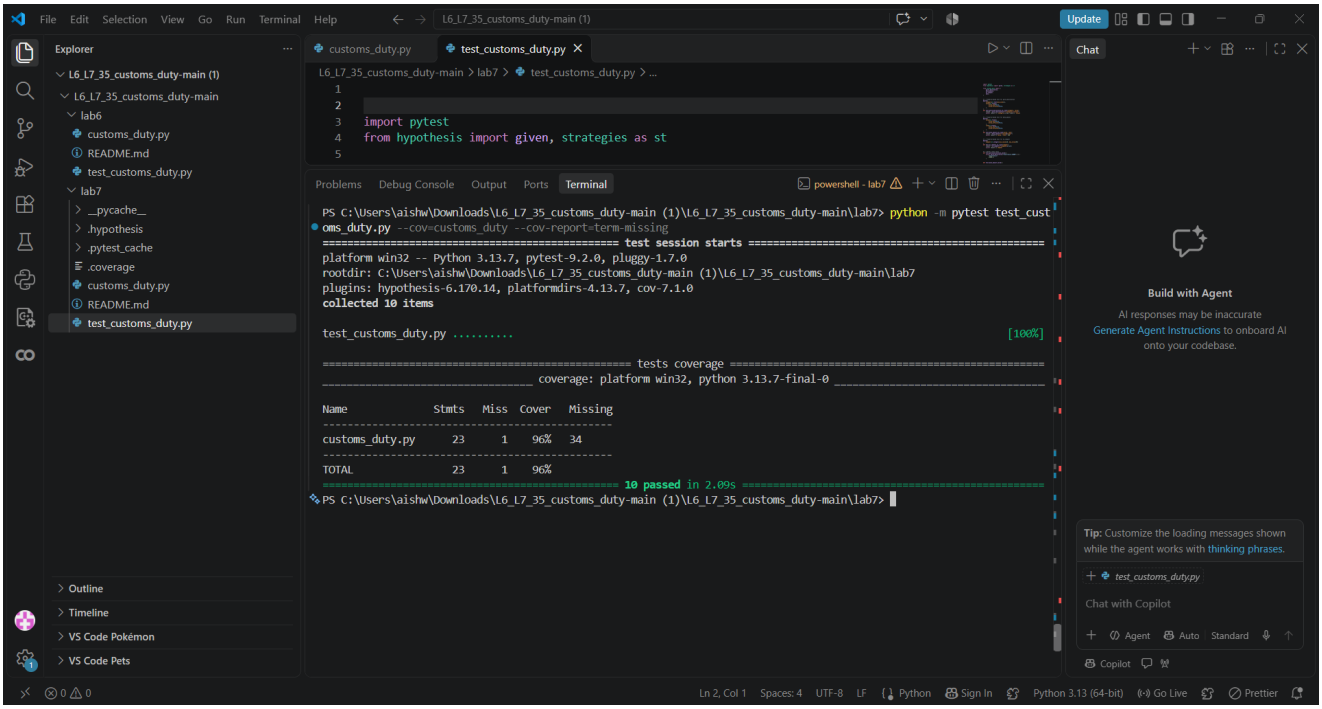
The parsing function was changed to split at the first colon using text.split(':', 1). The screenshot also shows the three initial function checks passing. This is evidence of the code correction and an intermediate retest, not a separate run of the full interactive application.

Evidence note: The available screenshots do not include a separate terminal screenshot of the corrected interactive application being run after the fix. This report therefore does not label this code-and-test screenshot as that application-output screenshot.



Screenshot 4 – Final Hypothesis test and coverage result

The final test command completed with 10 tests passed. The coverage output reports 96% coverage for customs_duty.py.



5. Result

The initial fuzz-testing run exposed a ValueError in parse_declaration() when the generated input contained more than one colon. The parsing logic was updated to use `text.split(':', 1)`, which limits the split to the first colon. The final test execution completed with **10 tests passed**. The displayed coverage report showed **96% coverage** for customs_duty.py.

6. Conclusion

This lab demonstrated how Hypothesis can generate unexpected inputs and reveal input-validation problems that may not be found through ordinary sample testing. The discovered parsing issue was corrected, and the final automated test run passed. The exercise improved understanding of property-based testing and defensive input handling.

7. Submission Checklist

Submit this PDF together with the corrected **customs_duty.py** application file and the updated **test_customs_duty.py** test file, as required by the lab handout.

Note: The lab handout asks for a screenshot of the fixed application output. That separate screenshot was not present in the screenshots available for this report; capture it by running `python customs_duty.py` after the fix if your instructor expects every step's terminal screenshot.